

Tuo Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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InfiniteSec

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Contents

1 About Cyfrin 2

2 About This Document 2

3 Disclaimer 2

4 Copyright and License 2

5 Audit Overview 3

6 Audit Scope 3

7 Risk Classification 4

8 Protocol Summary 4

9 Attestation 4

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Tuo on Tuo. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated September 28, 2026, was delivered to Tuo, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 5 days, the Cyfrin team conducted an audit on the Tuo code provided by Tuo. The findings consist of 3 High, 7 Medium & 10 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Tuo
Repository	tuo-app
Commit	aa54a8b70c73...
Fix Commit	7b892c674fa0...
Audit Timeline	Sep 10th - Sep 16th, 2026
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	3	3	0
Medium Risk	7	5	2
Low Risk	10	8	2
Informational	16	13	3
Gas Optimizations	4	3	1
Total	40	32	8

6 Audit Scope

The audit scope was limited to:

```
src/libraries/PerformanceFee.sol
src/libraries/PoolTwap.sol
src/libraries/PositionValuation.sol
src/libraries/TuoConstants.sol
src/libraries/TuoRoles.sol
src/TuoPositionNFT.sol
src/TuoSwapRouter.sol
src/TuoVault.sol
src/TuoVaultAdmin.sol
src/TuoVaultStorage.sol
src/TuoVaultViews.sol
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Tuo is an NFT-based yield platform on Arbitrum One with a Hyperliquid hedge leg: a customer deposits a supported token or native ETH, which the platform converts to USDC, deploys as Uniswap V3 concentrated liquidity in USDC-paired pools, and hedges with a short position on Hyperliquid. Each position is a transferable ERC-721 token. On-chain custody and per-position accounting are handled by a single `TuoVault`, which holds idle tokens and the Uniswap V3 LP NFTs, while the underlying LP assets reside in Uniswap pools.

- `TuoVault` is the custody and accounting core. It tracks each position's idle balances, Uniswap V3 LP NFTs, hedge margin, withdrawal state, and performance-fee basis.
- `TuoPositionNFT` mints and burns the ERC-721 position token.
- `TuoSwapRouter` executes the vault's token swaps through external swap aggregators.
- The Uniswap V3 integration holds LP positions through the NonfungiblePositionManager.
- The Hyperliquid hedge leg runs off-chain: an off-chain engine drives the hedging strategy, and margin is bridged between the vault and Hyperliquid.
- Supporting libraries compute pool TWAP prices, position valuations and performance fees.

9 Attestation

Cyfrin attests that it performed a security review of the Tuo code provided by Tuo.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-09-28-cyfrin-tuo-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Tuo has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=edfca17af5540b9fe724f7dc899a6cdc7295657cfabef1e081ce6cae65620216
F=2026-09-28-cyfrin-tuo-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.